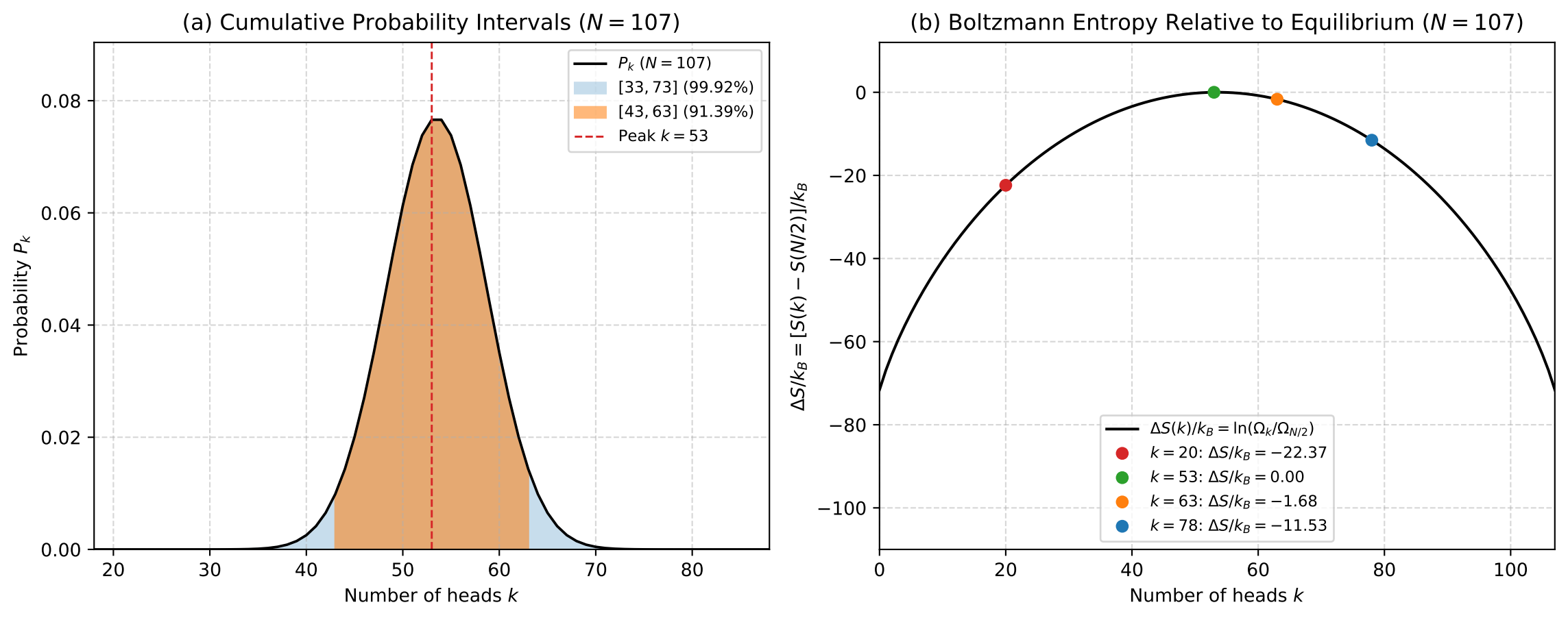
(a) Cumulative Probability Intervals ($N = 107$)
P_k ($N = 107$)
[33, 73] (99.92%)
[43, 63] (91.39%)
Peak $k = 53$
Probability P_k
Number of heads k
0.00
0.02
0.04
0.06
0.08
20
30
40
50
60
70
80
(b) Boltzmann Entropy Relative to Equilibrium ($N = 107$)
$\Delta S/k_B = [S(k) - S(N/2)]/k_B$
$\Delta S(k)/k_B = \ln(\Omega_k/\Omega_{N/2})$
$k = 20$: $\Delta S/k_B = -22.37$
$k = 53$: $\Delta S/k_B = 0.00$
$k = 63$: $\Delta S/k_B = -1.68$
$k = 78$: $\Delta S/k_B = -11.53$
Number of heads k
0
−20
−40
−60
−80
−100
0
20
40
60
80
100